1. What were the biggest challenges in implementing the prototype with the help of AI? How did you address them?

One of the biggest challenges was clearly communicating the requirements to the AI. Initially, the AI did not fully understand what I was asking for, which led to outputs that were incomplete or did not match the intended functionality. I had to be very specific and sometimes simplify or "break down" my requests into smaller steps. Additionally, I tested and ran the code multiple times, making adjustments and refining prompts until the prototype behaved as expected. Through this iterative process, I was able to guide the AI toward producing a more accurate result.

2. How well did the generated prototype align with your use cases, goals, and data models?

After several revisions, the generated prototype aligned well with the intended use cases, goals, and data models. Initially, there were mismatches between the design and implementation, but by refining prompts and making corrections, the final prototype reflected the core functionalities defined in the UML diagrams, such as searching for rooms, making reservations, and handling user interactions.

3. Which requirements or model elements were correctly implemented, and which were omitted or misinterpreted?

Some core requirements, such as searching for rooms and reserving a room, were implemented correctly early on. However, several important elements were initially missing or incorrect. For example, the login functionality was not always included, room capacity filtering did not work properly at first, and assistive features (such as accessibility options and language support) were either incomplete or misinterpreted. The AI tended to produce a basic version that met general requirements but did not fully match the detailed specifications. These issues were resolved by providing clearer instructions and refining the implementation step by step.

4. Would you recommend providing an AI agent with a requirements specification and UML artifacts to generate an application prototype? Why or why not?

Yes, I would recommend using an AI agent with requirements specifications and UML artifacts, but with some caution. The AI can be very helpful in quickly generating a starting point and reducing development time. It also helped me better understand how requirements translate into code. However, it is not perfect and requires clear instructions, careful review, and iterative refinement. Overall, it is a useful tool when combined with human guidance.

5. If you were to iterate on the prototype, what features or improvements would you prioritize, and why?

If I were to continue improving the prototype, I would prioritize enhancing the language feature. While a basic version was implemented, it did not fully translate the entire interface. Improving this would make the system more accessible and align better with the assistive features described in the design. Additionally, I would consider improving the user interface and adding more realistic functionality, such as a backend or database integration, to make the system more practical.